
A Proof Attempt Told the Search Where to Look: A Case Study in Machine-Assisted Mathematical Research

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We report a research loop in which a numerical search and a human proof attempt
2 corrected each other, in enough detail that the mechanism is visible. An adversarial
3 machine search over experimental designs suggested that a certain estimation-
4 theoretic lower bound could not be beaten even by an adversary who knows the
5 model’s functional form. The best ratio it found after thousands of designs was
6 1.005, so we wrote the bound down as structure-proof. The proof we then wrote
7 did not cover that claim. It established the bound only within two curvature lengths
8 of the anchor point, and the obstruction was a single inequality that fails outside
9 that region. Searching exactly where the proof stopped produced a counterexample:
10 a four-point design with one far probe of weight 3×10^{-4} , attaining ratio 0.8517,
11 verified at 50-digit precision. The earlier verdict was a search artifact, and the
12 result is now two-sided and exact. Neither half of the loop found this alone. We
13 place the episode inside a larger body of machine-checked mathematics on the cost
14 of self-knowledge in performative prediction, in which eleven measurement-forced
15 pivots are recorded in the code where each occurred.

16 1 Introduction

17 This paper is about how a piece of mathematics got made, and it uses one concrete body of theory as
18 the specimen. Our answer to the venue’s guiding question is narrow and, we hope, checkable. In
19 the loop we ran, machines were good at refuting and at certifying, people were good at proving, and
20 the interesting events happened where the two disagreed [Davies et al., 2021, Wagner, 2021]. We
21 document one such event completely, then ask how far it generalizes using ten more of its own kind.

22 The specimen is a theory of what it costs a decision-maker to learn how its own deployed actions
23 reshape the world it then retrains on. Section 2 says what that means in plain terms. What matters
24 here is that the theory is large enough to have a texture: proved theorems, numerically supported
25 conjectures, and a live simulation pipeline, each of which turned out capable of correcting the others.
26 Over its development, measurement forced eleven changes to claims that had already been written
27 down, and each pivot is recorded in the code file where it happened.

28 We lead with the eleventh, because it is the one where the two halves of the loop needed each other.
29 A machine search believed a statement. A human proof attempt failed to reach the statement, and
30 failed in a way that named the exact region where it stopped. A second machine search, aimed only
31 at that region, refuted the statement.

32 **What this paper is not.** It is not an autonomous mathematical agent [Hubert et al., 2025], and we
33 make no claim in that direction. The proofs here were written by people. What was automated is

numerical checking against declared tolerances, adversarial search over designs, and high-precision verification of candidate witnesses. Section 4 draws that line explicitly.

Contributions.

1. A reproducible account of one proof-and-search correction cycle, from the false belief through the failed proof to the frozen counterexample (Section 3). The resulting statement, Theorem 2, is two-sided and exact.
2. Ten further measurement-forced pivots from the same project, three of them the same pattern at smaller scale, and an account of what that sample does and does not support (Section 4).
3. The infrastructure that made the correction detectable: a theorem register carrying per-result proof status, preregistered numerical checks with tolerances declared before the run, and continuous integration over the derivation suite (Section 4), all shipped as an anonymized repository in the supplementary material.

2 The mathematical substrate, briefly

This section says what the theory is about, so that the counterexample in Section 3 has something to be a counterexample to. A reader who wants only the case study can take Theorem 1 on faith.

Consider a decision-maker whose deployed action changes the data it will later retrain on. A dealer who quotes a wide spread sees different order flow than one who quotes a tight spread, so the flow it must price is partly its own doing. This is performative prediction [Perdomo et al., 2020, Hardt and Mendler-Dünger, 2023]: retraining becomes a fixed-point iteration, and it converges to a point that is not the one maximizing value. Algorithms that close that gap [Izzo et al., 2021, Miller et al., 2021, Drusvyatskiy and Xiao, 2023] all need an estimate of how the distribution responds to the action, and they get it by perturbing deployments. Those perturbations are deployments of a real decision variable, so their cost lands in the decision-maker’s own objective.

Write h for the scalar action, h^* for the operating point the decision-maker would choose if it already knew everything, and $d_t = h_t - h^*$ for the deviation deployed at round t . Let ε be the response slope to be estimated, σ^2 the observation noise variance, Φ the value function, and γ_{PO} its curvature at h^* . The cost of exploring is the value forgone, $C_T = \sum_{t \leq T} \mathbb{E}[\Phi(h^*) - \Phi(h_t)]$. Formal statements are in Appendix B.

Theorem 1 (Exchange rate; T2). *Along any stationary exploration policy, for the least-squares slope estimator, $\text{Var}(\hat{\varepsilon}) \times C_T = \frac{1}{2} \gamma_{PO} \sigma^2 (1 + o(1))$, independent of the exploration amplitude, shape, schedule, and of how strongly the loop contracts.*

Learning how your own deployment moves the world costs a fixed amount. You can choose what to buy with it, but not how much to pay. The mechanism is a cancellation. Information accumulates as the centered energy of the design, cost accumulates as half the curvature times that same energy, and for mean-centered exploration the two coincide.

Three further results sit around that identity, all in Appendix B. No policy at all can beat the product, only match it (T3, T4). The cheapest way to spend a multidimensional budget is to probe where the objective is flat (T5a, C5.1). And a deployed agent can be made to correct only when a confidence sequence certifies that correcting cannot destabilize the retraining loop (L4, P7.1). The theory was also instantiated on a calibrated real-data problem and on a second economy, both in Table 1.

3 The lead story: a belief, a proof that missed it, and a witness

Theorem 1 is a statement about one estimator on one class of policies, and the minimax floors T3 and T4 extend it to all policies and all estimators. Both leave a natural question open. Suppose the adversary knows the functional form of the response, not just its slope. Concretely, suppose it knows that the response is $\tau_\theta(h) = C_0 + C_1 e^{-ch}$ and has only to learn the three parameters. Knowing the form is real information, so it might buy a design that beats the floor. Does it?

The machine’s answer, and why we believed it. We asked a search. For each candidate design μ the relevant quantity is the Cramér–Rao product normalized by the floor, $R(\mu) = (g^\top M(\mu)^{-1} g) \int (h - h^*)^2 d\mu$, where $M(\mu)$ is the design’s Fisher information and g is the gradient of the estimand. The

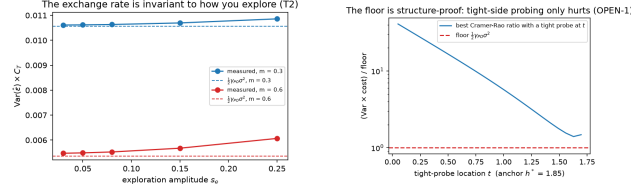


Figure 1: **Left:** the exchange rate of Theorem 1. Measured $\text{Var}(\hat{\varepsilon}) \times C_T$ against exploration amplitude at two contraction moduli, flat on the invariant (dashed). **Right:** the search that produced the false verdict. Best Cramér–Rao ratio with one tight probe at t , over the coarse weight grid with $w \geq 0.05$. On this family tight probes are monotonically worse, which is what we believed. The counterexample lives outside it: probe weight 3×10^{-4} , inside an asymmetric cluster the search never generated.

83 floor survives structural knowledge exactly when $R(\mu) \geq 1$ for every μ . Four thousand random
 84 designs plus 2,500 local refinements returned a minimum of $R = 1.0050$, and a systematic scan over
 85 designs pairing two near-anchor points with one distant probe found distant probes monotonically
 86 worse, reaching $R = 40.9$ at the grid’s furthest. Two independent strategies agreeing is the kind of
 87 evidence that gets written down, and we wrote it down: the floor is structure-proof, best attained ratio
 88 1.005.

89 **The proof did not reach the claim.** The proof attempt reduces $R(\mu)$ to a one-dimensional problem.
 90 A generalized Rayleigh identity gives $R(\mu) = \int d^2 d\mu / \inf \{ \int \phi^2 d\mu \}$, where $d(h) = h - h^*$ and
 91 the infimum runs over functions ϕ in the span of the model’s sensitivity components, normalized
 92 by $\phi'(h^*) = 1$. So it suffices to exhibit one such ϕ that is pointwise dominated by $|d|$. There is a
 93 canonical candidate, the unique element of the span with the right 2-jet at the anchor:

$$\phi_0(h) = \frac{2}{c} - e^{-c(h-h^*)} \left(\frac{2}{c} + h - h^* \right).$$

94 Writing $x = h - h^*$, the required domination $|\phi_0| \leq |x|$ holds for every $x \geq -2/c$, with equality
 95 only at $x = 0$ and $x = -2/c$. That is a clean elementary argument (Appendix C). It is also the whole
 96 story, because for $x < -2/c$ the domination does not merely resist proof. It is false, and it fails
 97 exponentially, since $\phi_0(x) = 2/c + e^{c|x|}(|x| - 2/c)$ there. The proof therefore delivered the floor on
 98 $[h^* - 2/c, \infty)$ and nowhere else, and it named the boundary.

99 **Searching where the proof stopped.** The failed step is not a gap in the argument. It is a location.
 100 We re-ran the adversarial search twice, once restricted to the reach and once with support allowed far
 101 below it. Inside the reach, a hardened search (three runs of 30,000 random designs plus 10,000 refine-
 102 ments) returned $R = 1.00054$, consistent with the proof and with an infimum of exactly 1. Allowed
 103 below the reach, the same search found $R = 0.85164$. The witness is a four-point design with support
 104 $\{1.8726, 1.8665, 1.3073, 0.0500\}$ and weights $\{0.003329, 0.955702, 0.040697, 0.000272\}$, in the
 105 register market with $c = 1.5$ and anchor $h^* = 1.8461$. Every support point except the one at 0.05 lies
 106 inside the reach, so that single below-reach probe of weight 3×10^{-4} is what breaks the floor. We
 107 froze the design and recomputed its ratio at 50-digit precision with `mpmath`, obtaining $0.851650\dots$;
 108 the check now runs on every push as V9.2.

109 **Why the first search missed it, exactly.** Two gaps, both now identified. The scan’s weight grid
 110 never went below 0.05, and the violating probe carries 3×10^{-4} . The random search did generate
 111 distant probes, but always alongside a symmetric near-anchor pair, and symmetric-pair-plus-far-probe
 112 families never break the floor: their ratio approaches 1 from above as the probe weight vanishes. The
 113 violation requires the asymmetric cluster. The earlier verdict was therefore neither noise nor a coding
 114 error. It was a correct summary of a search space that happened to exclude the answer, and nothing
 115 inside that search could have revealed the exclusion.

116 **Theorem 2** (Structure-proofness, exact reach; T9). (i) Every estimable design supported in $[h^* -$
 117 $2/c, \infty)$ satisfies $R(\mu) > 1$ strictly, and $\inf_{\mu} R(\mu) = 1$, approached but not attained. Within reach,
 118 knowing the family cannot beat the floor. (ii) The reach is exact: the four-point witness above attains
 119 $R = 0.8517$.

120 **The corrected result explains itself.** The two-sided statement says something the one-sided belief
 121 did not. Structure-proofness is an ignorance-of-curvature phenomenon with an exactly known extent.
 122 A design reaching far enough below the anchor learns the curvature parameter c cheaply enough to

bank that cheapness rather than pay for it, and the floor stops binding. Two boundary checks confirm the reading. With c known a priori, so that the ignorance is gone, the floor breaks outright and keeps falling with no limit the scan reaches ($R = 0.3002$ at probe $t = 1.0$, 0.0425 at $t = 0.05$, 0.0348 and still decreasing at $t = 0.001$). And the reach is exactly where a trust-region agent can deploy, since its clip is $r \leq 0.8 < 2/c = 1.333$: the constraint that keeps it safe is the one that keeps its lower bound valid. That was read off the corrected theorem, not designed.

4 The loop, and what it generalizes to

One episode is an anecdote. This section gives the sample it sits in and states what that sample supports.

The division of labour. Automated: numerical evaluation of every closed form against a simulation, at tolerances declared before the run; adversarial search over designs; high-precision arithmetic on frozen candidates; regression checking on every push. Not automated: every proof in this work, every choice of what to conjecture, and the decision to search below the reach. The machine in this loop is a fast, tireless, uncreative referee. It never proposed a theorem. It refuted several, and in the eleventh case only after a human argument told it where to look.

Eleven pivots, three of them the same shape. Measurement forced eleven changes to written-down claims, and three repeat the lead story at smaller scale. In E1, a saturation cap derived without noise failed by a factor of four against the noisy pipeline; the gap was a missing term, since retraining on noisy observations adds a stationary excitation floor of $(\sigma/\gamma)^2/(1 - m^2)$ per step, now itself verified within 0.2 to 6.4 percent. In E5, an injected misspecification ae^{-2ch} had decayed to about 10^{-3} by the operating point, so the anchored fit correctly never saw it; replacing it with a linear leak the family cannot represent produced the crossover the theory predicts. In E7, a claimed threefold advantage for shaped exploration collapsed to parity once the comparison arm was funded with the same energy, the unshaped arm having been given $r^2/3$ per step against the design's $2r^2/3$; the corrected claim is sharper, since the identification cliff is a support and amplitude phenomenon rather than a shape phenomenon. In each case the replacement was a stronger statement than the claim it displaced.

What the sample supports. A modest and specific thing: on this project, numerical falsification was cheap and frequent, and it improved claims rather than merely deleting them. It supports no claim about proof automation, about other areas of mathematics, or about what happens without a person deciding where to look. Unlike work in which search refutes a conjecture directly, here the search had already returned the wrong verdict, and a proof's failure rather than a larger budget redirected it. The eleventh pivot needed a proof to produce the region, and no further search inside the original space would have found it.

The infrastructure, which is the transferable part. A theorem register lists all 26 results with per-result proof status, distinguishing proved, proved for a stated class, and proved conditional on a hypothesis, so a numerically supported conjecture cannot quietly be cited as a theorem. A preregistered harness compares each measurement to a closed form at a tolerance fixed before the run; the state is 36 rows with 0 failures, one of them a deliberately out-of-scope cell reported as drift rather than excused. A derivation suite of 38 checks runs in continuous integration, and its first runs falsified three drafted claims, one with the wrong predicted sign. Every pivot is written into the code file where it occurred, which is why the eleventh survives as a record and not as a memory.

The honesty discipline is part of the contribution. We report the drift cell as drift. We label the sharp constant $\frac{1}{2}$ in T4 as numerically supported and open, since what is proved is the constant-factor version. And we say plainly that our own written verdict of structure-proofness was a search artifact. A record containing only successes cannot calibrate how much to trust the next numerical verdict, which is the only reason to keep one.

Limitations. Theorem 1 is local and its $o(1)$ is load-bearing: wide-side probing at finite amplitude undercuts the identity, and the out-of-scope saturating cell drifts +16 percent. The value-budget floor T4 is proved only to within a factor of 13.5 of the conjectured constant. Theorem 2(ii) lives inside the local-quadratic cost model, where under the true cost the same witness has ratio 1.0112, so the violation concerns the floor's own scope and not the underlying market. Which below-reach designs violate it, and by how much at most, is open. And this section rests on eleven pivots from one project by one group, which is a sample and not a study.

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A Related work

This appendix does what the four-page body has no room for: it places the episode of Section 3 against the work it resembles, and places the mathematical substrate of Section 2 against the literatures it draws on. Neither placement changes a claim in the body.

A.1 Machines and people in mathematics

The closest precedent for the shape of our loop is the use of computation to guide mathematical intuition toward statements people then prove [Davies et al., 2021], where a learned model’s saliency over a space of invariants pointed at the relationships that became theorems in knot theory and representation theory. The direction of travel there is machine to human: the computation supplies a candidate, the person supplies the proof. Our eleventh pivot runs the other way for one leg of the cycle, since the object the proof supplied was not a candidate statement but a region, and the region is what made the second search productive.

A second line lets search refute conjectures directly. Reinforcement learning over constructions has produced explicit counterexamples in extremal combinatorics and spectral graph theory [Wagner, 2021], and program search guided by a language model has produced both counterexample-like constructions and improved bounds [Romera-Paredes et al., 2024]. Our search is far less sophisticated than either, and that is the point of the episode: the first search was not underpowered in any way its own diagnostics could see. It was thorough over a space that excluded the answer, and it reported a minimum of 1.005 with two independent strategies agreeing. What told us the space was wrong was a proof that stopped at $h^* - 2/c$ and said so.

A third line automates proof itself, from olympiad geometry [Trinh et al., 2024] to olympiad-level formal reasoning with reinforcement learning [Hubert et al., 2025], on the infrastructure of large formal libraries [The mathlib Community, 2020]. We do none of this. Our proofs are written and checked by people and are not formalized, which is a limitation we state rather than defend. The older lineage of machine-assisted proof, from the four-colour theorem [Appel and Haken, 1977] to the formal Kepler proof [Hales et al., 2017], and the experimental-mathematics tradition of treating computation as evidence to be believed provisionally [Borwein and Bailey, 2008], is the tradition our verification harness sits in: it is bookkeeping about how much a numerical verdict is worth, not proof.

What we claim to add is narrow. It is a completely recorded instance of the failure direction, in which a numerical verdict was written down as a result, was wrong, and was corrected by a proof that could not reach it. The record includes the search that produced the false verdict, the reason it missed (a weight grid floored at 0.05 against a violating probe of 3×10^{-4} , and a symmetric-pair prior the violating design does not satisfy), the proof’s exact stopping point, and the frozen witness.

A.2 The substrate’s literatures

Performative prediction. The fixed-point framing and the gap between the stable point and the optimum are due to Perdomo et al. [2020], surveyed in Hardt and Mendler-Dünner [2023]. Algorithms that close the gap [Izzo et al., 2021, Miller et al., 2021, Drusvyatskiy and Xiao, 2023] and its stateful variant [Brown et al., 2022] all consume an estimate of the distribution’s response, obtained by perturbing deployments, and treat those perturbations as free. Jagadeesan et al. [2022] bound the regret of *finding* the performative optimum; the exchange rate prices *knowing your own feedback* instead, giving lower bounds and an invariance identity where they give algorithmic upper bounds. Bracale et al. [2025] design deployments to learn the distribution map and bound the regret of doing so at an order level; T2 prices the same activity exactly, in the objective’s own curvature. Lin and Zrnic [2024] show that misspecified structural models still help, which T7 sharpens into a budget-explicit rule.

Learning while earning. That myopic optimization can stall learning is classical in operations research, surveyed by den Boer [2015] and stated for dynamic pricing by Keskin and Zeevi [2014]. T1 is the performative version of that stall: it comes from the optimizer converging, not from weak feedback, and the cap is explicit. Lai and Robbins [1979]’s adaptive designs are points on the frontier T2 describes; what does not appear in that literature is the frontier itself, its floor, or a demand curve the decision-maker’s own actions generate.

301 **Dual control and experiment design.** Pricing an experiment inside the objective it serves is dual
302 control [Feldbaum, 1960] and least-costly identification [Bombois et al., 2006]. Here the plant is a
303 retraining learner, the budget is the optimizer’s own objective, and the constraint is loop contraction.
304 The lower bounds are van Trees arguments [Gill and Levit, 1995]. On the design side, task-optimal
305 exploration in linear systems [Wagenmaker et al., 2021] and the finding that naive exploration suffices
306 for the linear quadratic regulator [Simchowitz and Foster, 2020] sit against C5.1, which says isotropic
307 exploration overpays by an explicit curvature-dispersion factor once the design is priced in the
308 experimenter’s own objective. The classical canon [Kiefer and Wolfowitz, 1960, Pukelsheim, 2006]
309 supplies the A-optimality and Chebyshev-system machinery but never prices the design that way.

310 **The market.** The structural single-book market, its calibration, and the corporate-bond instantiation
311 behind the real-data row of Table 1 are those of REFLEX [Nagarajan and Ashok, 2026]. The quote-
312 summons-toxicity channel is the dealer’s adverse-selection problem [Guéant et al., 2013], and the fact
313 that a dealer’s own quotes feed the flow it then reads is the price-reading effect studied by Barzykin
314 et al. [2026].

315 B Assumptions and statements

316 This appendix collects the standing assumptions and the formal statements of every result the body
 317 refers to, and nothing else. Identifiers (L1, T2, T9, and so on) are the register identifiers used
 318 throughout the supplementary repository, so each statement here can be matched to its derivation
 319 document, its proof, and its numerical checks. The register itself, with per-result proof status, is
 320 `mlxor-derivations/THEOREMS.md`.

321 B.1 Standing assumptions

322 **A1** (local quadratic scope): Φ is C^3 and strongly concave on the operating interval, with expansions
 323 taken at h^* and third-order remainder bounded by $L_3 = \sup |\Phi'''|$. **A2** (exploration class): all
 324 policies are non-anticipating; A2-sym adds $\mathbb{E}[u_t | \mathcal{F}_t] = 0$, A2-adaptive is unrestricted, and costs are
 325 measured from the certainty-equivalent anchor. **A3** (noise exogeneity): the observation noise ζ_t is
 326 i.i.d., mean zero, variance σ^2 , independent of $\mathcal{F}_t$. A3' relaxes this to a feedback channel with gain ϕ .
 327 **A4** (trust region): $|d_t| \leq r$ pathwise. **A5** (stationary scale): $S_T = \Theta(T)$. **A6** (stable base loop): the
 328 retraining modulus satisfies $m < 1$.

329 The local dynamics are $d_{t+1} = -m d_t + u_t$ with $d_t = h_t - h^*$, observations $\tau_t = \tau(h^*) - \varepsilon d_t + \zeta_t$,
 330 and Φ satisfying $\Phi'(h^*) = \delta_0$ and $-\Phi'' = \gamma_{\text{PO}}$ at the operating point. S_{xx} denotes the centered
 331 design energy $\sum_{t \leq T} (d_t - \bar{d})^2$.

332 B.2 The exchange rate and its floors

333 **Lemma 1** (Cost equivalence; L1). *Under A1–A3 with conditionally symmetric exploration,*
 334 *the expected incremental performative cost satisfies $C_T = \frac{1}{2} \gamma_{\text{PO}} \mathbb{E}[\sum_{t \leq T} d_t^2] + R_T$ with*
 335 *$|R_T| \leq \frac{L_3}{6} \mathbb{E} \sum_t |d_t|^3$, and the realized cost's variance decomposes exactly as $\text{Var}(C_T^{\text{real}}) =$*
 336 *$\delta_0^2 \text{Var}(\sum_t d_t) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}(\sum_t d_t^2)$ with vanishing cross term.*

337 **Theorem 3** (Information saturation; T1). *With no deliberate exploration and $\rho(M) < 1$, the total*
 338 *design energy of the retraining trajectory equals $d_0^\top P d_0$, where P solves the discrete Lyapunov*
 339 *equation $P = I + M^\top P M$. Consequently the Fisher information for the response Jacobian is*
 340 *bounded in every direction, uniformly in the horizon. In the scalar case the cap is $d_0^2/(1 - m^2)$,*
 341 *strictly increasing in m : a fast-contracting loop reveals least about its own performativity (C1.1).*
 342 *Under the noisy feedback channel A3' the trajectory additionally carries a stationary excitation floor*
 343 *of $(\sigma/\gamma)^2/(1 - m^2)$ per step.*

344 The body's Theorem 1 is the asymptotic form of the following pathwise identity, which is what is
 345 proved in Appendix C.

346 **Theorem 4** (The exchange rate, pathwise; T2). *Under A1, A2-sym, A3 and A6, in the stationary*
 347 *regime, on every realization,*

$$\text{Var}(\hat{\varepsilon} \mid \text{design}) \times \frac{1}{2} \gamma_{\text{PO}} \sum_{t \leq T} d_t^2 = \frac{1}{2} \gamma_{\text{PO}} \sigma^2 \left(1 + \frac{T \bar{d}^2}{S_{xx}} \right) = \frac{1}{2} \gamma_{\text{PO}} \sigma^2 (1 + O_P(1/T)),$$

348 *independent of the exploration intensity, the horizon, and the modulus.*

349 **Theorem 5** (Minimax lower bound, deviation budget; T3). *Over all non-anticipating exploration*
 350 *policies with $\sum_{t \leq T} d_t^2 \leq D$ almost surely and all estimators, biased included, the minimax risk over*
 351 *an interval of width w satisfies $\inf \sup_{\varepsilon} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \sigma^2/(D + \sigma^2(\pi/w)^2)$, so the exchange-rate*
 352 *product is a floor, attained with equality by symmetric probing with least squares. The argument is a*
 353 *van Trees reduction in the style of Gill and Levit [1995].*

354 **Lemma 2** (Exploitation and information; L2). *For the symmetric two-point response family with*
 355 *half-width δ and any non-anticipating policy under the certainty-equivalent anchor, $\mathbb{E}[\text{gain}_T] \leq$*
 356 *$g_1 \delta^{3/2} T^{1/2} (\mathbb{E} S_T)^{3/4} \sigma^{-1/2}$ with $g_1 = \beta |h^* - \psi|$, via the identity $\mathbb{E}[s | \mathcal{F}_t] = \text{TV}(P_t^+, P_t^-)$*
 357 *and Pinsker's inequality. At the minimax scale this is an $O(T^{-1/2})$ fraction of the exploration cost.*

358 **Theorem 6** (Minimax lower bound, value budget; T4). *Under stationary-scale exploration and the*
 359 *certainty-equivalent anchor, for every non-anticipating policy and every estimator, $\sup \text{Var}(\hat{\varepsilon}) \times$*
 360 *$C_T \geq \gamma_{\text{PO}} \sigma^2 / 27 (1 - O(T^{-1/2}))$, the supremum over the two-point response family at the minimax*

scale, the constant arising from a Le Cam reduction optimized at $c = 2/3$. The sharp constant $\frac{1}{2}$, against which no simulated policy fell by more than Monte Carlo error, is conjectured and is part of the open problem OPEN-1. It is not proved here or anywhere in the repository.

B.3 Design and certification

Theorem 7 (A-optimal exploration; T5a). *Under the value budget $\text{tr}(\Gamma_{\text{PO}} M) \leq B$ on the exploration second moment M , the A-optimal design in the sense of the classical design theory of Kiefer and Wolfowitz [1960] and Pukelsheim [2006] is $M^* = (B / \text{tr} \Gamma_{\text{PO}}^{1/2}) \Gamma_{\text{PO}}^{-1/2}$, with value $(\text{tr} \Gamma_{\text{PO}}^{1/2})^2 / B$: explore where the objective is flat. Pricing the design in the experimenter’s own objective inverts the conclusion that naive exploration is optimal for the linear quadratic regulator [Simchowitz and Foster, 2020, Wagenmaker et al., 2021].*

Corollary 1 (Price of isotropic jitter; C5.1). *Isotropic exploration at matched budget overpays the A-optimal design by exactly the curvature-dispersion factor $F = d \text{tr}(\Gamma_{\text{PO}}) / (\text{tr} \Gamma_{\text{PO}}^{1/2})^2 \geq 1$, with equality if and only if all curvature directions are equal.*

Extended Chebyshev systems and their Fisher-information consequences are standard in the design literature [Pukelsheim, 2006]; the content below is that noiseless retraining trajectories fall on the wrong side of the support threshold.

Theorem 8 (Chebyshev unidentifiability; T6). *The sensitivity system of the structural family $\tau(h) = C_0 + C_1 e^{-ch}$ is an extended Chebyshev system. Any design with fewer than three support points has singular Fisher information, and since noiseless retraining trajectories have at most two effective support points, no retraining trajectory at any modulus identifies the response curvature c .*

Theorem 9 (Anchoring crossover; T7). *Under the value budget B and horizon T , the budget-and-horizon-optimal nonparametric secant estimator has $\text{MSE}_{\text{np}} = \gamma_{\text{PO}} \sigma^2 / (2B) + (\tau'''(h^*)/6)^2 (2B / (\gamma_{\text{PO}} T))^2$, so anchoring to a structural family with misspecification δ_{mis} wins, to leading order, which sharpens the message of Lin and Zrnic [2024] that misspecified structural models still help into a budget-explicit rule, if and only if $\delta_{\text{mis}} < |\tau'''(h^*)| B / (3 \gamma_{\text{PO}} T)$.*

Lemma 3 (Perturbed modulus; L4). *If the corrected update uses an estimated response with C^1 error e , the closed-loop modulus satisfies $|\hat{\rho} - \rho| \leq \eta \beta (\|e\|_\infty + |h^* - \psi| \|e'\|_\infty)$.*

Proposition 1 (Safety under pessimism; P7.1). *Given an anytime-valid confidence sequence for the response parameters at level $1 - \delta$, the pessimism rule, which keeps the worst-case modulus over the confidence set below $1 - c_{\text{margin}}$ and freezes the correction otherwise, keeps the realized modulus within the margin for all t simultaneously with probability at least $1 - \delta$. The result is conditional on the validity of the confidence sequence, and the register records it as such.*

B.4 Structure-proofness

Theorem 10 (Structure-proofness of the floor, exact reach; T9). *Fix the structural family $\tau_\theta(h) = C_0 + C_1 e^{-ch}$ ($C_1 > 0$, $c > 0$), the anchor $h^* > 0$, the sensitivity $s(h) = \partial_\theta \tau_\theta(h)$, and the estimand gradient $g = \partial_\theta \varepsilon(h^*) = -s'(h^*)$. For a finitely supported design μ define the normalized Cramér–Rao product*

$$R(\mu) = (g^\top M(\mu)^{-1} g) \int (h - h^*)^2 d\mu, \quad M(\mu) = \int s s^\top d\mu,$$

so that $R(\mu) \geq 1$ states that the family-knowing Cramér–Rao product $\text{Var}(\hat{\varepsilon}) \times C_T$ is at least the exchange-rate floor $\frac{1}{2} \gamma_{\text{PO}} \sigma^2$ under the local-quadratic (A1) cost, for unbiased estimators and, in the local-asymptotic-minimax sense, for all regular estimators. (i) Within reach the floor is structure-proof: if $\text{supp}(\mu) \subset [h^* - 2/c, \infty)$, then either $\varepsilon(h^*)$ is not estimable under μ , in which case the bound is infinite, or $R(\mu) \geq 1$, strictly whenever μ places mass off the two equality points $\{h^*, h^* - 2/c\}$, which is every estimable design, since a design supported on those two points alone leaves g outside the range of $M(\mu)$; and $\inf_\mu R(\mu) = 1$, approached but not attained by symmetric three-point designs collapsing at h^* . In particular the floor binds on every trust-region design with $r \leq 2/c$. (ii) The reach is exact: the pointwise domination underlying (i) fails strictly below $h^* - 2/c$, and an explicit four-point design with a single below-reach probe of weight 3×10^{-4} attains $R = 0.8517$, computer-verified at 50-digit precision as check V9.2.

409 C Proofs

410 This appendix proves the two results the body argues from: the exchange rate of Theorem 1, whose
 411 exact pathwise form is stated as Theorem 4, and the two-sided structure-proofness result of Theorem
 412 2, stated in full as Theorem 10. Both proofs are reproduced from the derivation repository without
 413 change; the check identifiers V2.1 and V9.1 to V9.4 refer to its verification suite. Proofs of the
 414 remaining statements in Appendix B are in that repository and are not reproduced here.

415 The cost-equivalence lemma

416 *Proof of Lemma L1 (cost equivalence).* Taylor's theorem with Lagrange remainder gives, for each t ,

$$\Phi(h^*) - \Phi(h_t) = -\delta_0 d_t + \frac{1}{2}\gamma_{\text{PO}} d_t^2 + r_t, \quad |r_t| \leq \frac{L_3}{6} |d_t|^3,$$

417 with $L_3 = \sup |\Phi'''|$ on the operating interval. Summing over $t \leq T$ and taking expectations, it
 418 remains to show $\mathbb{E}[\sum_t d_t] = 0$ under A2-sym. The recursion gives $\mathbb{E}[d_{t+1} | \mathcal{F}_t] = -m d_t + \mathbb{E}[u_t |$
 419 $\mathcal{F}_t] = -m d_t$, so $\mathbb{E}[d_{t+1}] = -m \mathbb{E}[d_t]$ by the tower property, and $\mathbb{E}[d_t] = (-m)^t \mathbb{E}[d_0] = 0$ when
 420 the loop starts at the operating point or in the stationary regime. The quadratic term contributes
 421 $\frac{1}{2}\gamma_{\text{PO}} \mathbb{E}[\sum_t d_t^2]$, which is the claim.

422 For the variance decomposition, write the realized cost (net of the cubic remainder) as $C =$
 423 $-\delta_0 X + \frac{1}{2}\gamma_{\text{PO}} Y$ with $X = \sum_t d_t$, $Y = \sum_t d_t^2$. Then $\text{Var}(C) = \delta_0^2 \text{Var}(X) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}(Y) -$
 424 $\delta_0 \gamma_{\text{PO}} \text{Cov}(X, Y)$. Under Gaussian innovations the vector $(d_1, \dots, d_T)$ is jointly Gaussian with
 425 mean zero, and every third moment of a zero-mean Gaussian vector vanishes; since $\text{Cov}(X, Y) =$
 426 $\sum_{t,s} \mathbb{E}[d_t d_s^2]$ is a sum of third moments, it is zero. (For non-Gaussian symmetric innovations the
 427 same conclusion follows from the sign symmetry $d \mapsto -d$ of the stationary law.) Verified: V0.1–V0.3,
 428 including the necessity check that an asymmetric rule breaks the expectation identity by exactly
 429 $-\delta_0 \mathbb{E}[\sum_t d_t]$. $\square$

430 The exchange rate

431 *Proof of Theorem T2 (pathwise exchange rate).* Under A3, conditionally on the design $\{d_t\}$ the ob-
 432 servations are independent with means linear in ε and variance σ^2 , so the least-squares slope has
 433 conditional variance $\text{Var}(\hat{\varepsilon} | \text{design}) = \sigma^2 / S_{xx}$ (Gauss–Markov). Multiplying by the quadratic
 434 cost $\frac{1}{2}\gamma_{\text{PO}} \sum_t d_t^2$ and using $\sum_t d_t^2 = S_{xx} + T\bar{d}^2$ gives the displayed identity exactly, on ev-
 435 ery realization. In the stationary regime $\bar{d} = O_P(T^{-1/2})$ by the stationary long-run variance
 436 $\text{Var}(\sum_{t \leq T} d_t) = Tv(1-m)/(1+m)(1+o(1))$, while $S_{xx}/T \rightarrow v$ a.s., so $T\bar{d}^2/S_{xx} = O_P(1/T)$.
 437 Verified: V2.1 (machine precision). $\square$

438 Structure-proofness, and the exactness of the reach

439 *Proof of Theorem T9 (structure-proofness within reach).* Write $x = h - h^*$, $d(h) = x$, and $V =$
 440 $\text{span}\{1, e^{-ch}, he^{-ch}\}$, the span of the components of s (a bijective correspondence $u \leftrightarrow \phi_u = u^\top s$
 441 between $\mathbb{R}^3$ and V , since $C_1 \neq 0$). Differentiating s gives $s'(h) = (0, -ce^{-ch}, C_1 e^{-ch}(ch - 1))^\top$,
 442 so $g = -s'(h^*)$ componentwise, which is the stated estimand gradient.

443 *Step 1: one-dimensional reduction.* For $M(\mu) \succ 0$ the generalized Rayleigh identity gives

$$g^\top M(\mu)^{-1} g = \sup_{u \neq 0} \frac{(u^\top g)^2}{u^\top M(\mu) u} = \sup_{\phi \in V} \frac{\phi'(h^*)^2}{\int \phi^2 d\mu} = \left(\inf \left\{ \int \phi^2 d\mu : \phi \in V, \phi'(h^*) = 1 \right\} \right)^{-1},$$

444 using $u^\top M u = \int \phi_u^2 d\mu$ and $u^\top g = -\phi'_u(h^*)$ (the sign is lost in the square). Hence

$$R(\mu) = \frac{\int d^2 d\mu}{\inf \{ \int \phi^2 d\mu : \phi \in V, \phi'(h^*) = 1 \}}.$$

445 *Step 2: the candidate and its pointwise domination.* Let

$$\phi_0(h) = \frac{2}{c} - e^{-cx} \left(\frac{2}{c} + x \right) = \frac{2}{c} \cdot 1 - e^{ch^*} \left(\frac{2}{c} - h^* \right) e^{-ch} - e^{ch^*} (he^{-ch}) \in V,$$

the unique element of V with 2-jet $(0, 1, 0)$ at h^* (the 2-jet map is invertible on V : in the basis $\{1, e^{-ch}, he^{-ch}\}$ its determinant is $c^2 e^{-2ch^*} \neq 0$); the expansion is $\phi_0(x) = x - (c^2/6)x^3 + O(x^4)$. Writing $f(x) = \frac{2}{c} - e^{-cx}(\frac{2}{c} + x)$ we claim

$$|f(x)| \leq |x| \quad \text{for all } x \geq -2/c, \quad \text{with equality iff } x \in \{0, -2/c\},$$

and $|f(x)| > |x|$ strictly for every $x < -2/c$. Indeed: (a) $v(x) = f(x) - x$ has $v(0) = 0$ and $v'(x) = e^{-cx}(1 + cx) - 1 < 0$ for $x \neq 0$ (since $1 + t < e^t$ for $t = cx \neq 0$); $v' \leq 0$ with equality only at the isolated point 0 makes v strictly decreasing on $\mathbb{R}$, so $f > x$ on $x < 0$ and $f < x$ on $x > 0$. (b) $f(0) = 0$ and $f'(x) = e^{-cx}(1 + cx) > 0$ for $x \geq 0$, so $f \geq 0$ on $[0, \infty)$ and, with (a), $0 \leq f \leq x$ there. (c) $q(x) = f(x) + x$ has $q(-2/c) = 0 = q(0)$ and $q'(x) = e^{-cx}(1 + cx) + 1$, where $r(x) = e^{-cx}(1 + cx)$ is strictly increasing on $x < 0$ ($r'(x) = -c^2 x e^{-cx} > 0$) from $r(-2/c) = -e^2 < -1$ to $r(0) = 1$; so q' changes sign exactly once on $(-2/c, 0)$, from negative to positive, and $q \leq 0$ there with equality only at the endpoints; with (a) ($f \geq x$, equality only at 0) this gives $x \leq f \leq -x$ on $[-2/c, 0]$. (d) For $x < -2/c$, $r(x) < -e^2$ gives $q' < 0$, so moving left from $-2/c$ the function q increases above 0: $f(x) > -x = |x| > 0$, and in fact $f(x) = 2/c + e^{c|x|}(|x| - 2/c)$ grows exponentially.

Step 3: the bound. If $\text{supp}(\mu) \subset [h^* - 2/c, \infty)$, Step 2 gives $\int \phi_0^2 d\mu \leq \int d^2 d\mu$, so by Step 1, $R(\mu) \geq \int d^2 d\mu / \int \phi_0^2 d\mu \geq 1$. Strictness: equality in Step 2 holds only at $x \in \{0, -2/c\}$, so any μ placing positive mass off $\{h^*, h^* - 2/c\}$ (in particular any with three distinct support points) has $\int \phi_0^2 d\mu < \int d^2 d\mu$ and $R(\mu) > 1$. If $M(\mu)$ is singular, two cases: $g \notin \text{range } M(\mu)$ makes $\varepsilon(h^*)$ non-estimable (infinite bound); $g \in \text{range } M(\mu)$, which does occur for special two-point designs (a one-parameter family of in-reach pairs; their finite pseudo-inverse products were checked to obey the bound, e.g. $R = 1.063$ at the pair $\{1.627, 2.127\}$), keeps the sup-representation valid over $\{u : u^\top M u > 0\}$, and $\int \phi_0^2 d\mu = 0$ would put $u_0 \in \text{null } M$ with $u_0^\top g = -1 \neq 0$, contradicting $g \in \text{range } M$; so $\int \phi_0^2 d\mu > 0$ and the bound goes through verbatim. Throughout, the Cramér–Rao functional bounds unbiased estimation, and all regular estimators in the local-asymptotic-minimax sense.

Step 4: the infimum is 1. The Cramér–Rao functional is invariant under C^1 reparametrization, and the 2-jet map $\theta \mapsto (\tau_\theta(h^*), \varepsilon(h^*), \tau''_\theta(h^*))$ is a local diffeomorphism: its Jacobian $[s(h^*) - s'(h^*) \quad s''(h^*)]$ has determinant $C_1 c^2 e^{-2ch^*} \neq 0$ (direct computation; V9.4). In jet coordinates η the model reads $\tau = \eta_1 - \eta_2 x + \frac{1}{2} \eta_3 x^2 + O(x^3)$ with sensitivity $\tilde{s}(x) = (1, -x, x^2/2)^\top + O(x^3)$, and the estimand is η_2 . For the symmetric three-point design μ_δ uniform on $\{h^* - \delta, h^*, h^* + \delta\}$, scale by $D = \text{diag}(1, \delta, \delta^2)$: $M_\eta(\mu_\delta) = D \hat{H}_\delta D$ with $\hat{H}_\delta \rightarrow \hat{H}_0$ invertible (the exact quadratic-model moment matrix), so $(M_\eta^{-1})_{22} = \delta^{-2} (\hat{H}_\delta^{-1})_{22}$ and $R(\mu_\delta) \rightarrow R_{\text{quad}}$. In the exact quadratic model, for any symmetric design (odd moments $m_1 = m_3 = 0$), the $(2, 2)$ cofactor gives $(M_\eta^{-1})_{22} = 1/m_2$ and $R_{\text{quad}} = m_2 \cdot (1/m_2) = 1$. Hence $R(\mu_\delta) \rightarrow 1$ (measured rate $R(\mu_\delta) - 1 = O(\delta^2)$; V9.2), the infimum is 1, and by Step 3 it is not attained.

Step 5: the reach is exact. Below $h^* - 2/c$ the domination of Step 2 fails strictly, and the failure is realized by a design, not only by the candidate: the frozen four-point witness (support $\{1.8726, 1.8665, 1.3073, 0.0500\}$, weights $\{0.003329, 0.955702, 0.040697, 0.000272\}$ after normalization, in the register market $c = 1.5$, frozen anchor $h^* = 1.8461$) has $R = 0.851650 \dots$, verified at 50-digit precision (mpmath) and reproduced in float64 by `run_open1.py` (V9.2). All its support except the probe at 0.05 lies within reach: one vanishing-weight below-reach probe breaks the floor. Symmetric pair-plus-far-probe families do *not* break it (their ratio approaches 1 from above as the probe weight vanishes); the violation requires the asymmetric cluster, which is why coarse searches missed it. Verified: V9.1–V9.4. $\square$

490 D Verification rows

Table 1: Selected rows from the preregistered check harness. The full table (36 rows, 0 FAIL), all seeds, tolerances, and code are in the supplementary repository. The drift row is a deliberately out-of-scope cell, measured and reported rather than excused.

Claim (register)	Measured	Predicted	Verdict
Exchange rate, $m=0.3$, two amplitudes (T2)	0.01062, 0.01070	0.01056	pass
Exchange rate, $m=0.6$, two amplitudes (T2)	0.00549, 0.00569	0.00535	pass
Saturating environment (outside A1)	+16% over rate	out of scope	drift
Feedback floor, $m=0.3/0.6/0.85$ (T1 corollary)	within 0.2–6.4%	$(\sigma/\gamma)^2/(1-m^2)$	pass
Design arm $\text{sd}(\hat{c})$; jitter/design ratio (T6)	0.234; $7.8\times$	Fisher 0.248; ≥ 3	pass
Certified agent reaches h_{PO} ; trust region (L4)	1.258; 0 violations	1.296; 0	pass
Isotropic/A-optimal risk ratio, $d=8$ (T5a)	1.529	$F = 1.506$	pass
Nonparametric MSE at (B, T) optimum (T7)	0.0132	0.0123	pass
Second economy, 4 cells (T2, exact in LQ); agent	0.384, 0.192; 12.65	0.384, 0.192; 12.5	pass
Multi-book shaped/isotropic steady-state risk (T5a)	1.20 (flat control 1.00)	$> 1.15 (\approx 1)$	pass
Real-data calibration port, 10 cells	10/10 reproduce	within 0.8%	pass
Within-reach min CR ratio; witness beyond (T9)	1.0005; 0.8517	≥ 1 ; < 1	pass

491 E Reproducibility and scope of this version

492 **What ships.** The supplementary repository contains the derivation documents and the theorem register
 493 (26 results, 38 numerical derivation checks), the simulation pipeline (environments, estimators,
 494 designs, agents, baselines; 36 measured-versus-predicted rows; 9 unit tests), the real-data port with its
 495 10-cell validation, and the eleven-pivot record in the pipeline README. Both verification suites run
 496 in continuous integration on every push in the project repository; the two workflow files are not part
 497 of this anonymized copy. The structural market used throughout, and the corporate-bond calibration
 498 behind the real-data row of Table 1, are those of REFLEX [Nagarajan and Ashok, 2026], on which
 499 this work builds. All results are deterministic from seeds, numpy-only, and CPU-only. Reported final
 500 errors are 300-step tail averages over the deployed path, stable within 0.01 under horizon perturbation,
 501 with a worst measured spread of 0.0062.

502 **What is stated here but not proved here.** This version proves Theorem 1 and Theorem 2, the two
 503 results the body argues from. The minimax floors and the design and certification results (T3, T4,
 504 T5a, C5.1, T6, T7, L4, P7.1) are stated in Appendix B and proved in the supplementary repository, in
 505 `mlxor-derivations/latex/proofs.tex` and the corresponding derivation documents, with the
 506 register recording each result’s proof status.

507 **Provenance of the numbers.** Every quantity in the body and in Table 1 is copied from a committed
 508 artifact: `results/RESULTS.md` for the harness rows, `results/OPEN1.md` for the reach, the witness,
 509 and the search history, `results/REALDATA.md` for the calibration port, `results/STABILITY.md`
 510 for the tail-average stability, `results/full_run.log` for quantities not surfaced in the summary
 511 tables, and `THEOREMS.md` for the register. The runs behind them were executed and recorded on 1
 512 September 2026. This version re-narrates them and does not re-run them, since none of its changes
 513 touch their inputs.